

HPS Gaussian-Process Resonance Search

Analysis Note v4.9.12: Expected-Limit Bands and Observed Diagnostics

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September 5, 2026

Cumulative 300-toy stage. This results-only note uses 300 background-only toys per mass (toy IDs 0–299). This is the planned full cumulative ensemble; all earlier toy IDs are retained exactly. All bands and empirical limit-tail diagnostics are pointwise and conditional on frozen GP states.

1 Inputs and construction

The seven scopes are the exact final-sample curves used in the most recent Harvard results section: full 2015, full 2016, the optimized-support 2021 10% sample, their three pairs, and the all-three shared- ϵ^2 combination. The 2021 configuration uses the frozen 36–300 MeV GP support and factor-15 resolution-scaled length ceiling. No 2021 1% or 2016 10% comparison enters these results.

At each mass, a latent background vector is drawn from the conditioned mass-local GP prediction and then Poisson sampled. The same dataset draw is reused by every scope containing that dataset at a fixed mass and toy ID, so the standalone and combined comparisons are paired. GP hyperparameters are not refit inside toys. Each pseudoexperiment is evaluated with the same bounded, piecewise-asymptotic 90% CL_s construction as the observed v4.9.12 curve. For numerical stability, fixed-strength nuisance fits minimize a data-constant-centered Poisson deviance; the fitted likelihood, statistic, and root acceptance gates are unchanged from the parent solver. During the 100-toy continuation, an inherited free-strength optimizer status failed at one deterministic coordinate despite a finite candidate. The contracted solver now uses the same likelihood-equivalent centering as a retry only when a raw free fit is unsuccessful. The released ensemble records 2 centered-profile retry activations (1 free-profile and 1 null-profile); every retry still passes the original likelihood-nesting and root gates, and no toy is dropped or reseeded.

The bands are empirical 2.5%, 16%, 50%, 84%, and 97.5% quantiles. Because different mass points are generated independently, these are pointwise expected-limit bands rather than a scan-wide ensemble. They do not establish unconditional coverage, toy-calibrated CL_s , a look-elsewhere correction, or global significance. Full-2016 and 2016-inclusive curves also retain the previously disclosed cross-process GP-state replay exception.

2 Results

Figure 1 supplies the requested standalone total-window view. It always selects the combination containing every final dataset available at that mass; it does not select among scopes according to the observed or expected limit. Thus the stitching rule is fixed by dataset support rather than by a favorable fluctuation. The displayed 2021 pieces use the optimized-support 10% configuration.

Figure 2 is the direct Brazil-band analogue of the earlier v4.2/v4.5 combination display, now evaluated on the current final inputs. Local motion in the outer envelope should still be read with the finite-ensemble granularity of the current 300-toy stage in mind.

Final observed limit with expected bands over the total search window

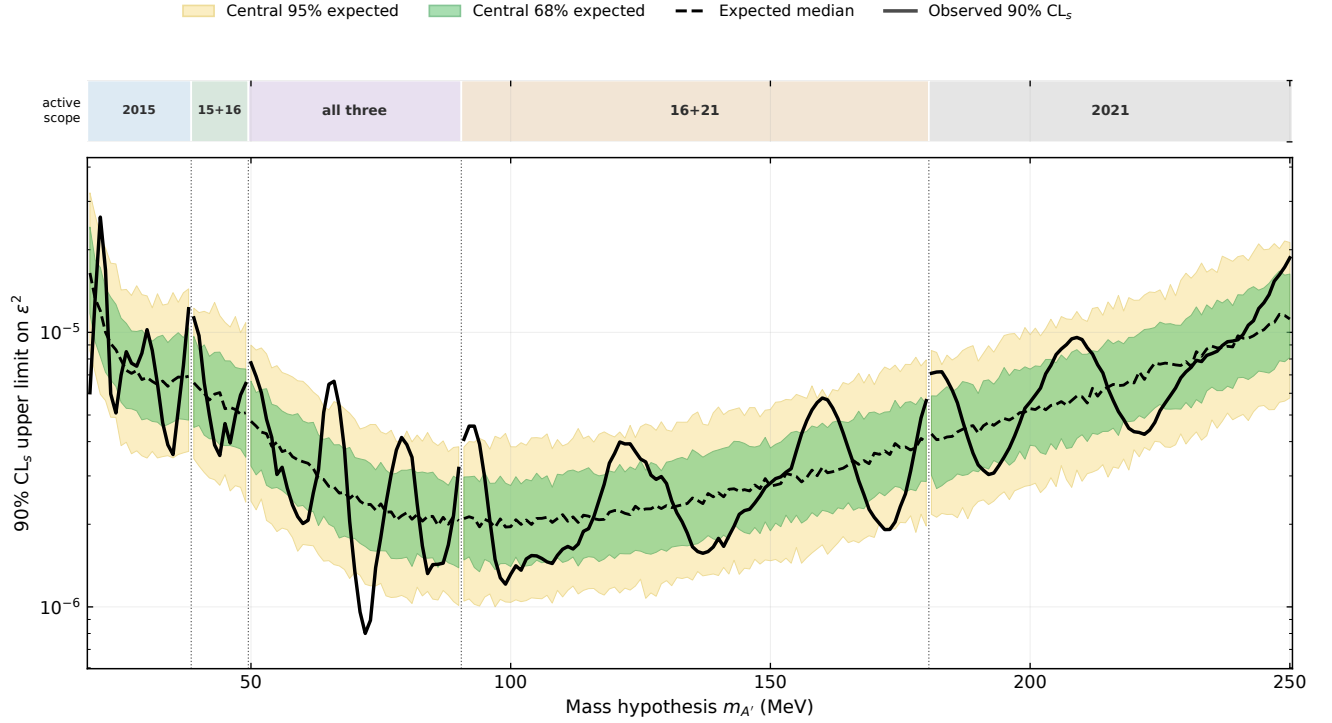


Figure 1: The “final” mass-by-mass maximal-available-dataset observed limit and expected bands from 300 background-only toys per mass over the total search window, 19–250 MeV. The bands are pointwise and conditional on the frozen GP states. The scope strip states which final datasets enter each interval: 2015 at 19–38 MeV, 2015+2016 at 39–49 MeV, all three at 50–90 MeV, 2016+2021 at 91–180 MeV, and 2021 at 181–250 MeV. Scope boundaries are explicit because this stitched display is not one invariant likelihood across the whole range.

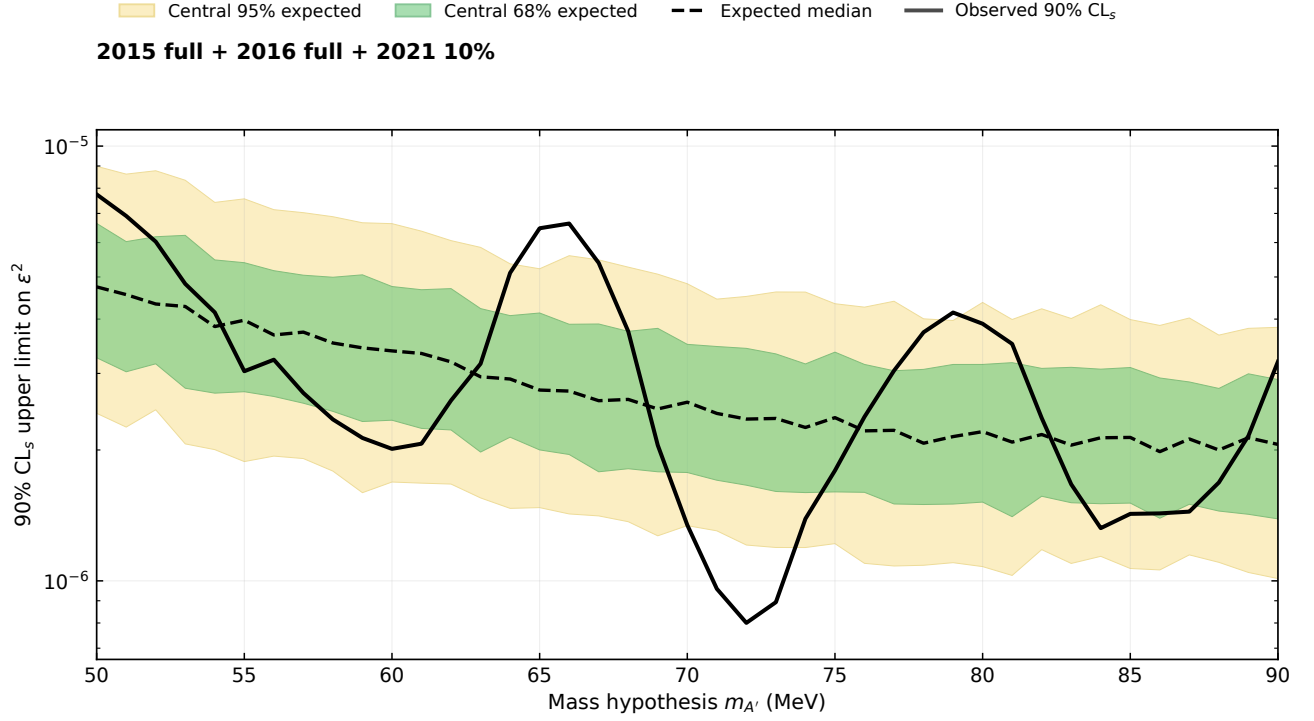


Figure 2: Observed and expected 90% CL_s upper limit for the all-three shared-coupling combination, using 300 background-only toys per mass. Yellow and green show the central 95% and 68% pointwise background-only ranges; the dashed curve is the expected median. The expected ranges are conditional on the frozen GP states.

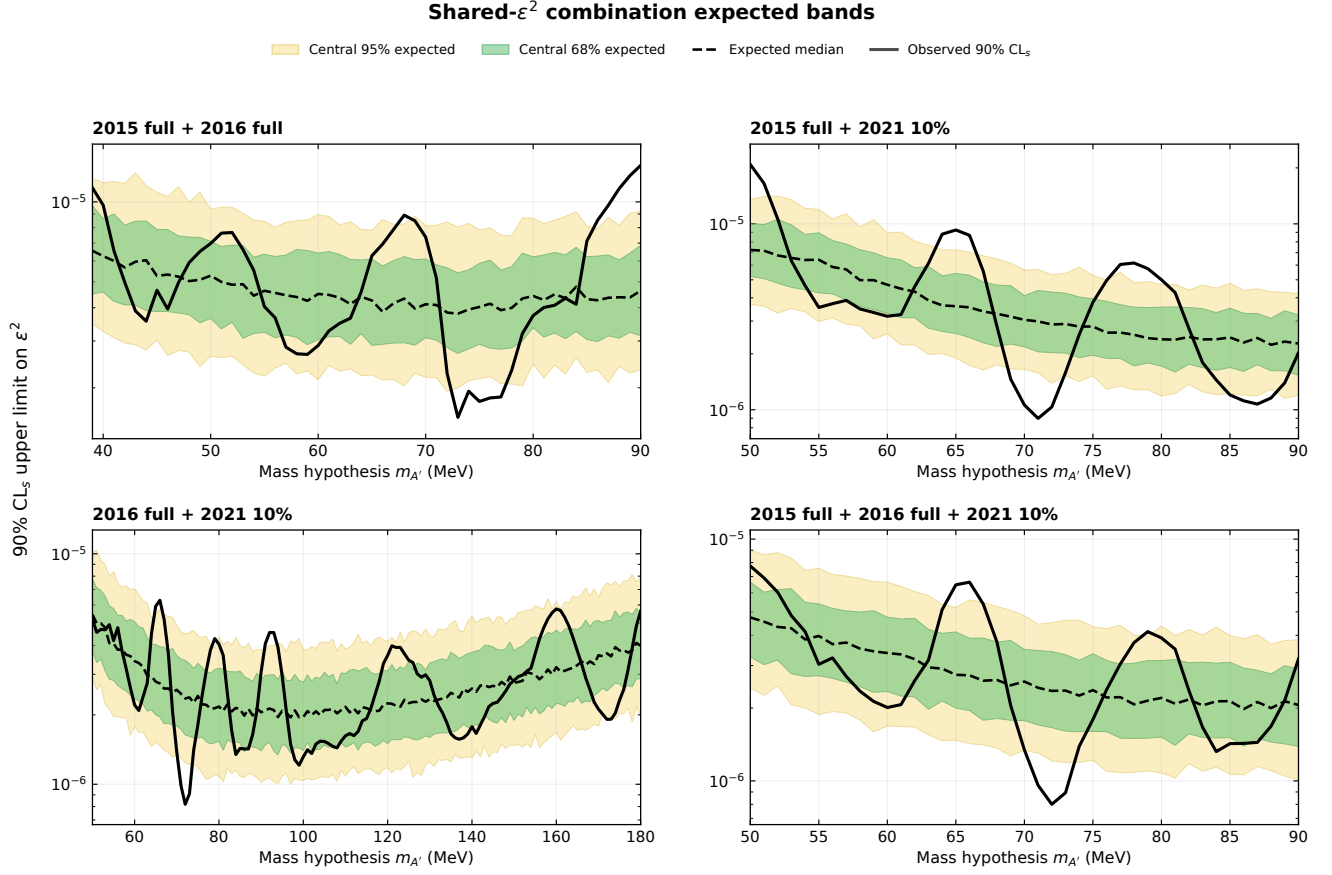


Figure 3: Pairwise and all-three shared- ϵ^2 limits with 300 background-only toys per mass. Each panel uses the same constituent pseudo-observations at a given mass and toy ID. Bands show the conditional central 68% and 95% ranges.

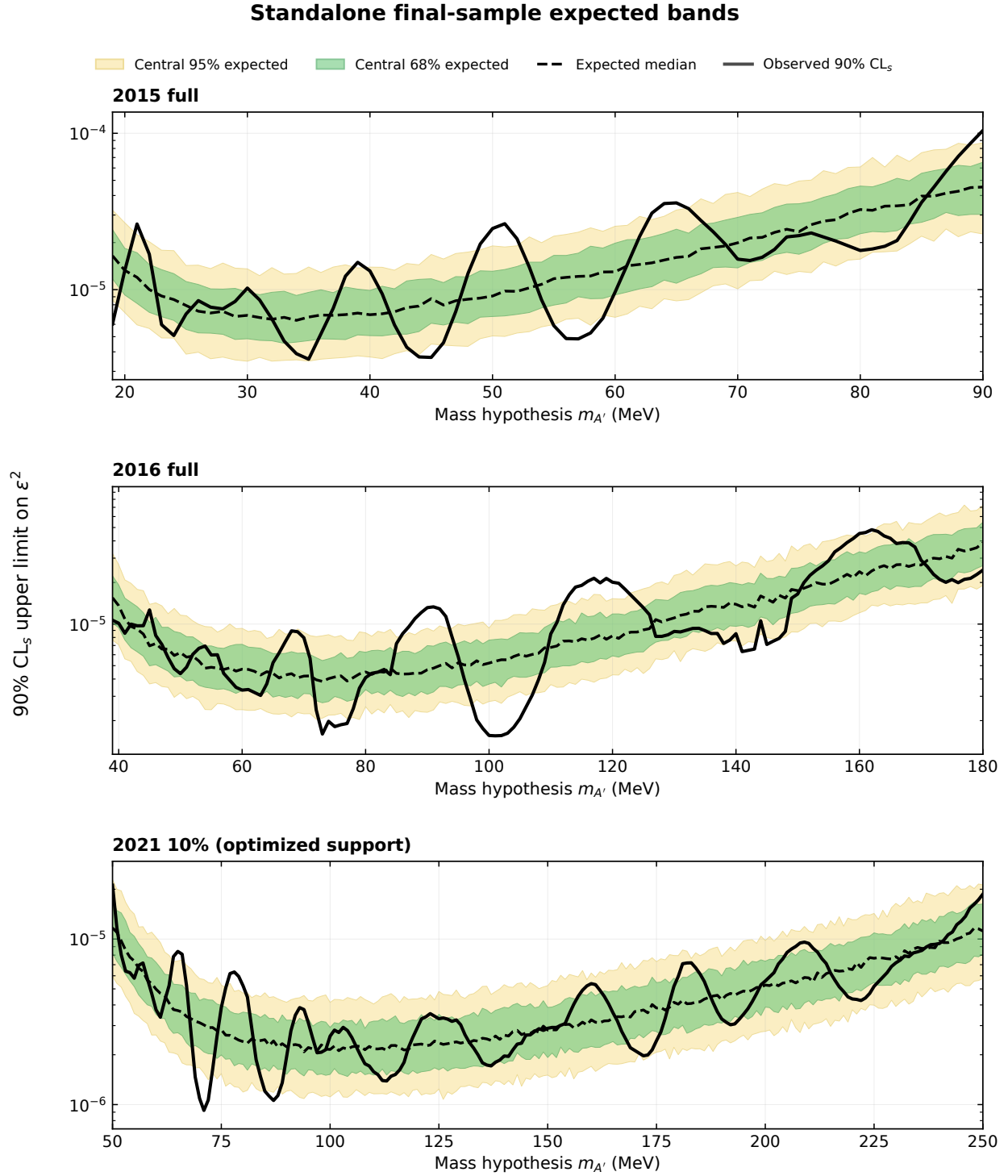


Figure 4: Individual 2015 full, 2016 full, and 2021 10% observed limits with 300 background-only toys per mass. Bands show the conditional central 68% and 95% ranges. The 2021 panel uses the optimized native-10% configuration with 36–300 MeV GP support.

2.1 Observed-limit and analytic p-values

Two different diagnostic families are reported and must not be conflated. For an observed upper limit $U_{\text{obs}}(m)$ and toy limits $U_t(m)$ at a fixed mass,

$$p_{\text{strong}}(m) = \frac{1}{N} \sum_t \mathbf{1}[U_t(m) \leq U_{\text{obs}}(m)], \quad (1)$$

$$p_{\text{weak}}(m) = \frac{1}{N} \sum_t \mathbf{1}[U_t(m) \geq U_{\text{obs}}(m)], \quad (2)$$

$$p_{\text{two}}(m) = \min[1, 2 \min(p_{\text{strong}}, p_{\text{weak}})]. \quad (3)$$

A small p_{strong} marks an unusually tight observed limit, while a small p_{weak} marks an unusually loose observed limit. The two-sided number is a symmetric reviewer diagnostic. These are upper-limit ensemble positions, not discovery p-values and not coverage measurements. The one-sided fractions have granularity $1/N = 1/300 \simeq 0.00333$, while p_{two} is constructed from them; zero counts are retained as zero in the CSV and shown at half a count only for logarithmic plotting.

The analytic p_0 is separate: it is the frozen observed curve’s one-sided, fixed-mass, asymptotic profile-likelihood background-only p-value. It is not derived from these 300 band toys. The local values themselves have no trials correction. Section 4 gives separate analytic scan-level references.

Combination limit-tail diagnostics and analytic local discovery p-value

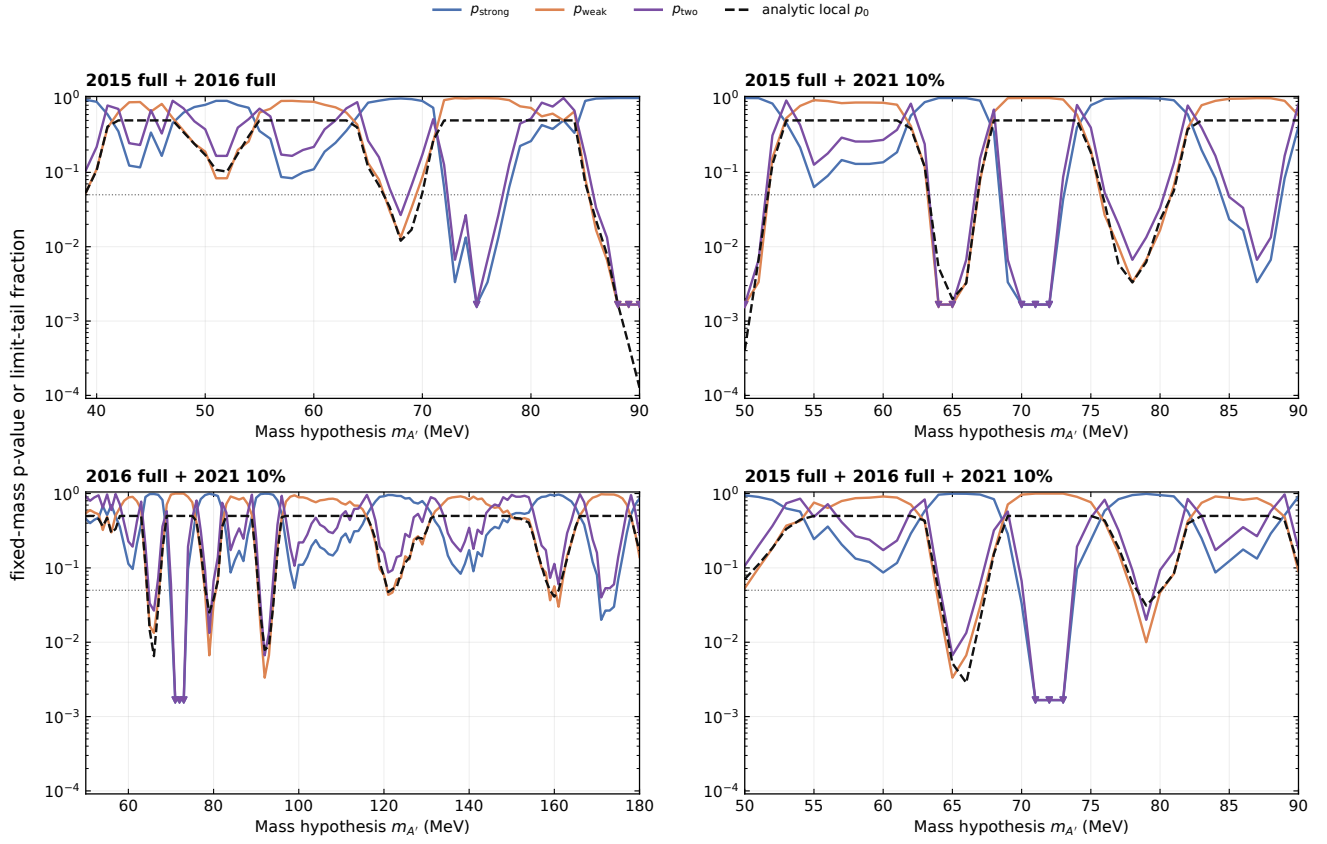


Figure 5: For each shared- ϵ^2 combination, the three empirical observed-limit tail diagnostics from 300 background-only toys per mass, alongside the analytic local discovery p_0 . A downward triangle at $1/(2 \times 300)$ denotes exactly zero toys in that tail. The plotted marker is only a display position; the recorded count remains zero. The dotted line marks 0.05.

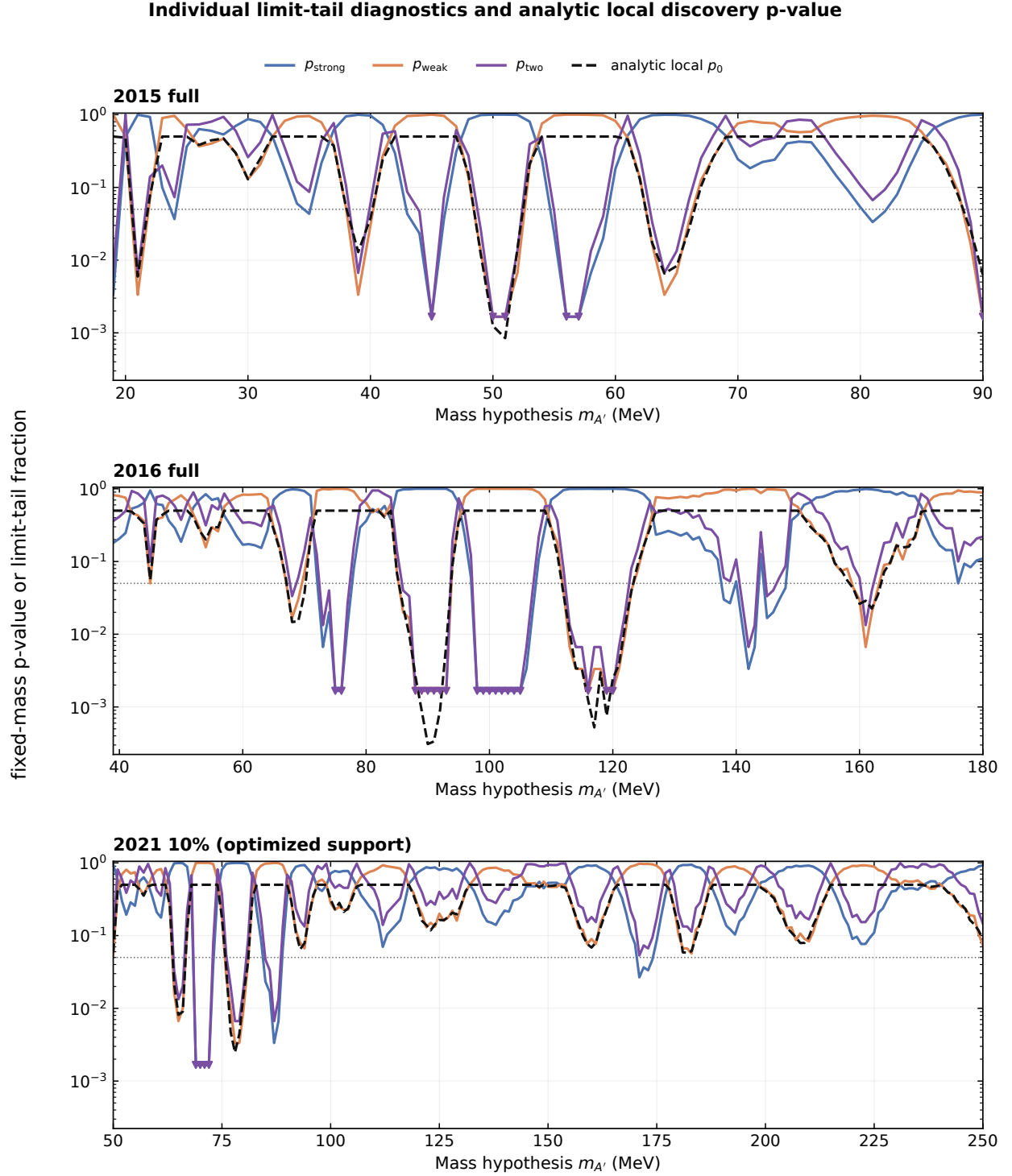


Figure 6: Individual final-sample limit-tail diagnostics using 300 background-only toys per mass and the analytic local p_0 . Definitions, the zero-count marker at $1/(2 \times 300)$, and the dotted 0.05 reference are the same as in Figure 5.

Scope	m [MeV]	p_{strong}	p_{weak}	p_{two}	analytic p_0	Z_{local}
2015 full	51	1	0	0	0.000 846	3.139
2016 full	90	1	0	0	0.000 309	3.424
2021 10%	78	0.997	0.003 33	0.006 67	0.002 47	2.810
2015 + 2016	90	1	0	0	0.000 127	3.659
2015 + 2021	50	1	0	0	0.000 403	3.351
2016 + 2021	66	0.987	0.0133	0.0267	0.0065	2.484
All three	66	0.993	0.006 67	0.0133	0.002 84	2.765

Table 1: Fixed-mass diagnostics at each scope’s smallest analytic local p_0 . The one-sided empirical fractions have granularity $1/300 \simeq 0.00333$ and p_{two} is constructed from them; the selected minimum analytic p_0 is a compact reporting choice, not a trials-corrected global test.

Table 1 gives explicit values at the smallest analytic local p_0 in each scope. These entries retain their fixed-mass interpretation. Complete values for all 680 scope–mass coordinates are in the machine-readable p-value ledger.

2.2 Compact limit summary

Table 2 reports, for each scope, the mass at which the 300-toy expected median is smallest. These minima summarize the plotted curves; they are not optimization inputs and do not select the analysis card.

Scope	m [MeV]	Observed	-2σ	Median	$+2\sigma$
2015 full	34	3.893×10^{-6}	3.554×10^{-6}	6.360×10^{-6}	1.269×10^{-5}
2016 full	73	1.600×10^{-6}	2.085×10^{-6}	3.847×10^{-6}	8.220×10^{-6}
2021 10%	94	3.750×10^{-6}	1.087×10^{-6}	2.084×10^{-6}	4.558×10^{-6}
2015 + 2016	73	1.548×10^{-6}	1.999×10^{-6}	3.800×10^{-6}	8.031×10^{-6}
2015 + 2021	88	1.157×10^{-6}	1.292×10^{-6}	2.236×10^{-6}	4.309×10^{-6}
2016 + 2021	97	1.489×10^{-6}	1.001×10^{-6}	1.949×10^{-6}	4.183×10^{-6}
All three	86	1.429×10^{-6}	1.060×10^{-6}	1.982×10^{-6}	3.868×10^{-6}

Table 2: Observed and conditional expected ϵ^2 limits at each scope’s minimum expected-median mass. The outer columns are empirical central-95% endpoints from 300 toys per mass.

Across all 680 scope–mass coordinates, all 204,000 toy limits were finite and closed at $CL_s=0.10$ within the frozen numerical tolerance. No latent GP draw used the clipping fallback. The exact per-toy limits and pseudo-observation hashes are retained in machine-readable ledgers.

3 Reproducibility and continuation

The ensemble is cumulative. The contracted cumulative schedule is complete at 300 toys per mass. Atomic per-mass checkpoints make the continuation resumable. The contract binds the analysis card, GP states, observed curves, solver, and protocol to the saved toys.

4 Equivalent global and trials-adjusted p-values

For each stated search window, let $p_{\min}$ be the smallest analytic local p_0 among its M tested masses. The Šidák equivalent global reference and the Bonferroni trials adjustment are

$$p_{\text{Šidák}} = 1 - (1 - p_{\min})^{N_{\text{eff}}}, \quad (4)$$

$$p_{\text{Bonferroni}} = \min(1, Mp_{\min}). \quad (5)$$

The Šidák expression is exact for independent uniform null p-values when the exponent is the number of tests. Here N_{eff} is a resolution-spacing approximation, following the earlier analysis-note convention. At ordered masses m_i , set σ_i to the smallest mass resolution among the datasets entering that scope. We use the fixed spacing convention $W = 2.25$ and

$$N_{\text{eff}} = \max\left(1, \min\left[M, \sum_{i=1}^{M-1} \frac{m_{i+1} - m_i}{W(\sigma_i + \sigma_{i+1})/2}\right]\right).$$

The frozen analysis card supplies every resolution, including the total-window composition changes. The resulting Šidák value approximates a scan-global probability with an effective count that has not been calibrated with scan toys.

The Bonferroni column uses every tested grid point and gives a conservative trials-adjusted value without requiring independence [1]. Its guarantee assumes valid local p-values; here those local inputs are conditional asymptotic results and retain the disclosed 2016 numerical exception. The empirical p_{strong} , p_{weak} , and p_{two} are not inputs to either correction.

Scope	Window [MeV]	M	N_{eff}	$m_{\min}$ [MeV]	local p_0	$p_{\text{Šidák}}$	$p_{\text{Bonferroni}}$
2015 full	19–90	72	13.624	51	0.000 846	0.0115	0.0609
2016 full	39–180	142	16.062	90	0.000 309	0.004 95	0.0439
2021 10%	50–250	201	26.402	78	0.002 47	0.0633	0.497
2015 + 2016	39–90	52	8.878	90	0.000 127	0.001 12	0.006 58
2015 + 2021	50–90	41	8.165	50	0.000 403	0.003 29	0.0165
2016 + 2021	50–180	131	20.708	66	0.0065	0.126	0.851
All three	50–90	41	8.165	66	0.002 84	0.023	0.117
Total window	19–250	232	35.381	66	0.002 84	0.0959	0.66

Table 3: Equivalent global and trials-adjusted references for each stated window. The Šidák column uses a resolution-based effective trials count; Bonferroni uses all M tested grid points. Both use the frozen analytic local p_0 , independently of the 300 band toys.

For the fixed total 19–250 MeV search, the minimum remains at 66 MeV, with local $p_0 = 0.00284$. The resolution model gives $N_{\text{eff}} = 35.3814$ and $p_{\text{Šidák}} = 0.0959$ ($Z_{\text{Šidák}} = 1.305$). Using all 232 tested masses gives $p_{\text{Bonferroni}} = 0.66$.

Each row corrects its own mass window. Treating all seven displayed scopes as one family gives 680 tests and $p_{\text{Bonferroni}} = 0.0861$, using that family’s own minimum local p_0 .

A scan-toy trials p-value is unavailable from this ensemble because the 300 band toys are generated independently at each mass. A calibrated scan-tail calculation requires coherent whole-spectrum pseudoexperiments and the maximum discovery statistic over each scan, or a separately validated upcrossing calculation [2].

References

- [1] SAS/STAT User’s Guide, The MULTTEST Procedure, “p-Value Adjustments.”
- [2] E. Gross and O. Vitells, “Trial factors for the look elsewhere effect in high energy physics,” Eur. Phys. J. C **70**, 525–530 (2010), arXiv:1005.1891.